

Masoud Kishani Farahani

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Education

- **M.Sc. Program in Energy Systems Engineering, Sharif University of Technology, Department of Energy Engineering, GPA (17.11/20)**  **Oct. 2020 - Feb. 2023**
 - **Thesis:** Fault detection and diagnosis in refrigeration systems using machine learning algorithms.
- **B.Sc. in Physics Engineering, Faculty of Science, Arak University**  **Sep. 2016 - Jan. 2020**

Research Interests

- Fault Detection and Diagnosis & Digital Twin
- Data-Driven Predictive Maintenance
- Data-Driven Energy Modeling
- Interpretable Machine Learning
- Smart Grid & Smart Cities
- Optimal design & operation of energy systems

Publications

- **M. K. Farahani, MH. Yazdi, M. Talaei, A. Rajabi, "Enhancing Energy Efficiency in Supermarkets: A Data-Driven Approach for Fault Detection and Diagnosis in CO2 Refrigeration Systems", Applied Energy, 2024.**
- **M. K. Farahani, M. Kolivandi, A. Rajabi, M. Talaei, "Enhancing Fault Detection in CO2 Refrigeration Systems: Optimal Sensor Selection and Robustness Analysis Using Tree-Based Machine Learning", arXiv:2503.17694, 2025.**
- **M. K. Farahani, M. Kolivandi and A. Rajabi-Ghahnavieh, "Comparative Analysis of Energy Efficiency Evaluation Methods for Buildings: ISO 13790 vs. EnergyPlus Simulation*", 9th International Conference on Technology and Energy Management (ICTEM), 2024, doi: [10.1109/ICTEM60690.2024.10631972](https://doi.org/10.1109/ICTEM60690.2024.10631972).**
- **M. Kolivandi, M. K. Farahani, A. Rajabi-Ghahnavieh, "Probabilistic Energy Routing Algorithm Enabling Reliability Assessment in Local Energy Internet", (ICTEM), 2024, doi: [10.1109/ICTEM60690.2024.10631934](https://doi.org/10.1109/ICTEM60690.2024.10631934).**
- **M. K. Farahani, A. Rajabi, Two weeks of electrical energy dataset at a five-minute sampling rate from supermarket buildings of a chain store in Iran, 2022.**

Extra-Curricular Involvement

- **Reviewer of *Elsevier Journal of CO2 Utilization*.**  **2025 - Present**
 - Reviewed papers related to Machine Learning and Energy Systems.

Research-based Projects

- Development of a **Linear Programming** Code Based on the **Simplex Method** Revised in **Python**, Advanced Mathematical Programming Course. 📅 2020
 - **Role:** Researcher **Supervisor:** Dr. Ramin Roshandel
- **Energy optimization** in smart buildings **using mixed-integer linear programming using GAMS software**, Energy Systems Modeling Course. 📅 2021
 - **Role:** Researcher **Supervisor:** Dr. Hossein Khajepour
- Predicting **Air Quality Index** with Tree-based Machine Learning Techniques: A Case Study in Ray City, Tehran Province, Energy and Climate Change Course. 📅 2021
 - **Role:** Researcher **Supervisor:** Dr. Hossein Khajepour
 - Summer **Energy Management Experience** in Sharif Department of Energy Engineering, Using **EnergyPlus**, Energy consumption management in the building Course. 📅 2021
 - **Role:** Researcher **Supervisor:** Dr. Abbas Rajabi-Ghahnavieh

Work Experience and Employment

- **Senior Manager, Environmental Energy Audit Group**, Alternative to Mandatory Military Service, Institute of Standards and Industrial Research of Iran (INSO), Tehran, Iran. 📅 2023 - Present
 - **Main Project:** Pioneered the development of a groundbreaking framework and methodology for the environmental assessment of polluting companies, a first-of-its-kind initiative for the National Standard Organization of Iran.
- **Supervisor of Master Thesis**, Guided and supervised a student in the university's laboratory, Department of Energy Engineering, Sharif University of Technology, Tehran, Iran. 📅 2021 - 2023
 - **Thesis: Fault Detection and Diagnosis in Air-Handling Units Using Virtual Sensors, Optimal Sensor Set, and Robustness Analysis.** Student: Mohammad Hamzeloo.
 - **Supervisor:** Dr. Abbas Rajabi-Ghahnavieh
- **Teaching Assistant for Energy Management in Buildings Course**, Department of Energy Engineering, Sharif University of Technology, Tehran, Iran. 📅 2021 - 2022
 - **Supervisor:** Dr. Abbas Rajabi-Ghahnavieh

Academic Experience and Certifications

- **Machine Learning Specializaion** Comprises of 3 courses, Supervised Machine Learning: Regression and Classification, Advanced Learning Algorithms (implemented deep learning models including MLP, LSTM, Transformers and U-Net from scratch), Unsupervised Learning, Recommenders, Reinforcement Learning (Coursera Certificate)

- **IBM Data Science Professional Certificate** Hands-on training in statistics, data visualization, machine learning, and deep learning, including Python, SQL, Jupyter Notebooks, GitHub, RStudio, Pandas, NumPy, Scikit-learn, Matplotlib, and more. (IBM Certificate)
- **Deep Reinforcement Learning in Python** (DataCamp)
- **Object-Oriented Programming in Python** (DataCamp)
- **Energy Modeling** Gained deep understanding of analytical optimization methods and their applications in energy systems (18.5/20, M.Sc.)
- **Energy Management in Buildings** Provided a deep understanding of calculating and evaluating building energy demands and achieving optimal indoor environments (**19/20**, M.Sc.)

Language Skills

- Previous TOEFL Test: **88** | Reading: **24**, Listening: **21**, Speaking: **21**, Writing: **22**  3 Feb. 2025

Skills & Qualifications

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| • Python: NumPy, Pandas, Polars, Matplotlib | • Machine Learning: Scikit-learn, Keras, Tensorflow |
| • EnergyPlus , EES, PVsyst | • Software: L ^A T _E X, Microsoft Azure, Microsoft Office |
| • Optimization: GAMS, Pyomo | • Git, Linux, PostgreSQL, FastAPI |
| • MATLAB, C++ | • Arduino, Arduino Cloud, ESP8266 |

Honors & Awards

- **First Place Poster Award**, 9th International Conference Energy Management  Mar. 2024
- Ranked **43rd (Top 0.08%)** in the national M.Sc. entrance exam.  Sep. 2020
- Selected as a qualified participant in the 2nd Scientific Physics Olympiad for University Students.  Oct. 2020
- Ranked **1st** in the Photonics Entrance Exam.  Sep. 2020

References

- **Dr. Abbas Rajabi-Ghahnavieh** - Assistant Professor, Energy Engineering Department, Sharif University of Technology, Tehran, Iran rajabi@sharif.edu (Google Scholar)
- **Dr. Amin Faraji Astaneh** - Assistant Professor, Departments of Physics, Sharif University of Technology, Tehran, Iran faraji@sharif.ir (Google Scholar)